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Energy Efficient Compact Approximate Multiplier for Error-Resilient Applications

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***Abstract*—The primary goal of approximate computing is enhancing system performance, such as energy efficiency, speed, and form factor. Despite the growing use of approximate multipliers, the design of efficient approximate compressors — a fundamental multiplier block — remains a significant challenge. In this paper, 8-transistor and 14-transistor 4:2 compressors are proposed. Both compressors exploit CMOS technology and a constant and conditional approximation of selected inputs, exhibiting fewer negative errors. As a result, a resource-expensive error recovery module is eliminated, yielding superior performance as compared with prior art. The 14-transistor architecture yields a lower error rate compared to the 8-transistor architecture, trading off lower area for higher accuracy. The compressor-tailored circuit architecture is also proposed and evaluated using image multiplication. The proposed multiplier exhibits 50% area savings and 93% lower power-delay-product compared to the exact multiplier, as well as higher accuracy, and 38% PDP enhancement compared with the state-of-the-art.**

***Index Terms*—Approximate computing, Compressor, Multiplier, Image Multiplication.**

I. Introduction

APPROXIMATE computing is a promising approach for enhancing the speed and efficiency of arithmetic circuits [1], [2]. This emerging concept has been employed across various error-resilient domains to balance accuracy and circuit performance in error-tolerant applications, such as image processing. Approximate computing aims to enhance circuit performance by disregarding certain inputs/outputs at a reasonable expense of the output accuracy.

Compressor circuit is a fundamental component in more complex circuits like multipliers [3]–[9]. Various compressor application-specific architectures have been proposed, offering distinct advantages and limitations [7], [9]. Compared to exact compressors that consider all the inputs and outputs, certain inputs and outputs are discarded with approximate compressors [10]. Owing to substantial reduction in circuit complexity, input/output counts, and load capacitance, these circuits generally outperform exact circuits [11], [12]. Compressors and full-adders (FAs) are fundamental blocks of a multiplier. Thus, compact approximate multipliers are typically constructed using alternative compressors and FAs comprising a smaller number of transistors, as part of the partial product reduction tree (PPRT) of multiplier circuit. Note that the large area of existing approximate solutions results in higher power and delays. Therefore, saving the error rate and circuit parameters remains a crucial challenge in compressors.

To tackle this pressing issue, we propose a novel circuit based on majority logic (ML) [7]. Compared to the 12-transistor cell in [7], our proposed approximate compressor uses eight transistors within the approximate PPRT sub-circuit of an 8-bit multiplier. To reduce the error rate of our first proposed compressor, we propose a more accurate 14-transistor compressor. We investigate the efficiency of our proposed circuits based on the system-wide performance of an approximate multiplier during image multiplication. Using our circuits, the number of partial products (PPs) and exact compressors are significantly reduced resulting in elimination of 15 AND gates from the PPRT stage. Only one exact compressor is used *totaling 786 transistors — a 49% and 93%* reduction in area and energy consumption, respectively, compared to exact multiplier. Based on a comprehensive analysis of existing approximate compressors, each proposed and state-of-the-art compressors exhibits a distinct Boolean function and circuit architecture, trading off accuracy, power, and area in a unique manner.

The paper is organized as follows. Section II reviews the existing approximate compressors. The proposed compressors and multiplier are described in Section III. Section IV provides the simulations results while Section V concludes the paper.

II. Related Work

In [3], [4], two approximate 4:2 compressors are proposed, prioritizing circuit efficiency over accuracy with reduced delay. The multiplier in [13] exhibits higher error rates and increased area. An approximate compressor in [5] computes error distances (ED) using a hash table and mitigates errors with an error recovery module (ERM), yielding high accuracy and area consumption. In [6], a configurable multiplier with exact and approximate modes is proposed, exhibiting high area and up to a 50% output errors. Majority logic (ML) [7] and CMOS-based [8] circuits significantly reduce area and power consumption at the expense of a higher error rate. Circuit-stacking-based approximate compressors [9], algorithm-based optimization with error control modules [11], and probability-based error correction [12] techniques show reduced error while requiring a larger area.

III. Proposed Approximate Compressors

Majority logic (ML), a robust and fault-tolerant digital logic design paradigm, leverages the power of collective decision-making to enhance reliability, simplicity, and efficiency in electronic circuits. An ML gate functions through an odd-input logic principle, generating a True (False) output when more than half of its inputs are True (False). Due to the

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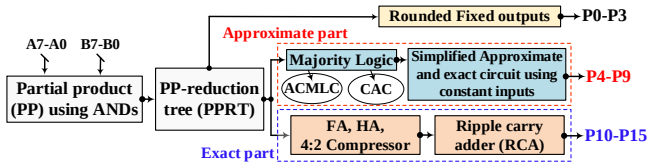

Fig. 1. Proposed design flow of 8-bit multiplier.

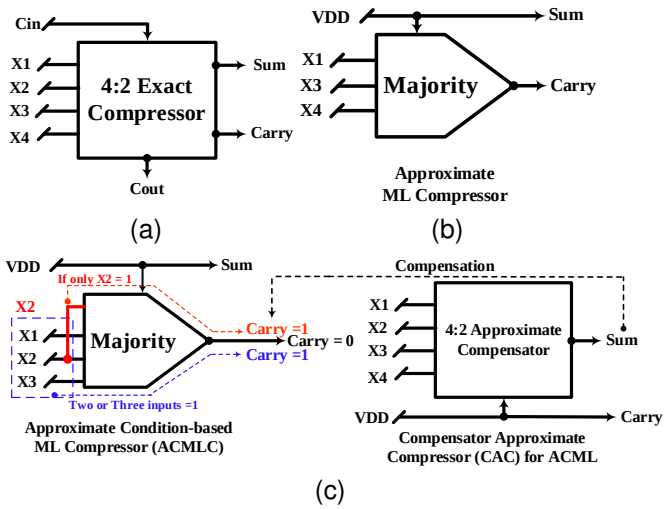

Fig. 2. Schematics of the compressor circuits (a): Exact. (b): ML-based, and (c): ACMLC and CAC, respectively.

increased configurability, ML-based circuits typically require fewer gates. Hence, the design of ML-based compressors is a promising approach for reducing the circuit area without sacrificing accuracy. Fig. 1 shows our design flow for an 8-bit multiplier to produce a combination of fixed, approximate, and exact outputs. The ML-based approximate components (c.f., Section III-A and III-B) generate the approximated bits and transfer them to the simplified circuits consisting of half adder (HA) and full adders (FAs) with at least one fixed input.

A. 4:2 Approximate Condition-based ML Compressor

The ML circuit architecture described in [7] is utilized and improved with a two-stage approximation approach. Fig. 2 shows the circuit schematics, including the cell view of the exact compressor circuit (c.f., Fig. 2a), ML-based compressor (c.f., Fig. 2b, [7]), and the proposed approximate condition-based ML (ACMLC) compressor and compensator approximate compressor (CAC) for the ACMLC (c.f., Fig. 2c). Note that the ML-based compressor [7] considers only three out of four inputs (input X2 is ignored) and a fixed output ($Sum = V_{DD}$). The $Carry$ output is True if at least two of the inputs are '1' (*i.e.*, X1 + X3 + X4 equals two or three), yielding a total of four negative and four positive errors.

Extensive research has recently been conducted on the impact of negative and positive errors in approximate circuits [14]. In the approximate region of a multiplier, each approximate compressor generates an error, leading to a significant error distance in the columns. Employing an error recovery module (ERM) based on specific input patterns for individual approximate compressors is crucial for minimizing overall error distance and reducing negative errors. Consequently, the importance of ERM circuits for mitigation of negative errors has been emphasized [5], [11], [12].

TABLE I
TRUTH TABLE FOR THE ACMLC COMPRESSOR.

X4	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1
X3	0	0	0	0	1	1	1	1	0	0	0	0	1	1	1	1	1
X2	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1	1
X1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	1
Carry	0	0	1	1	0	1	1	1	0	0	1	1	0	1	1	1	1
Sum	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Error Distance (ED)	+1	0	+2	+1	0	+1	+1	0	0	-1	+1	0	-1	0	0	-1	

Employing a multiplier that utilizes approximate compressors intelligently, owing to their low negative error rates and consistent error generation, eliminates the necessity for ERM. Accordingly, we propose the ACMLC circuit architecture, as shown in Fig. 2c, to simultaneously mitigate negative error and reduce the circuit area. In addition to the typical majority-based approximation (*i.e.*, one of three inputs, X4, is ignored and Sum is set to V_{DD}), the $Carry$ output is '1' in the following two scenarios: (i) two or three of the inputs are '1' (similar to [7]) as highlighted in blue in Fig. 2c, or (ii), only X2 is '1' (and the rest of the inputs are '0'), as highlighted in red in Fig. 2c. For all other input combinations, the $Carry$ output is '0', yielding $Sum_{ACMLC} = V_{DD}$, $Carry_{ACMLC} = (X1 \cdot X3) + X2$. Table I shows the Truth Table for the ACMLC 4:2 compressor, yielding a total of 9 errors (only one error more than [7]), of which only one output exhibits an error distance of two (ED = 2) and only 3 errors are negative (compared to 4 negative errors of [7]).

Furthermore, our architecture results in a significant simplification of the Boolean functionality and reduction in the circuit area, as shown in Section IV. Compressor architectures that exhibit a higher number of negative errors, a compensator circuit is needed to enable the data flow between the approximate and exact sub-circuits of a multiplier (see [5], [11], [12]). To address this challenge, we propose CAC circuit architecture (c.f., Fig. 2c) for ACMLC compensation. The architecture is based on a typical 4:2 compressor architecture with a fixed carry output, $Carry = V_{DD}$ and approximate Sum, $Sum = (X1 \cdot X2)(X3 + X4) + (X3 \cdot X4)$.

With this approach, there are only seven errors (all with error distance of ED = 1), and one negative error. Alternatively, the number of transistors is increased from eight in ACMLC to 14 in CAC. The simplicity of the proposed ACMLC compressor as compared to the ML-based compressor [7] and CAC naturally leads to lower area. Although CAC has two more transistors than [7], it benefits significantly from less negative errors. Based on unique features of the ACMLC and CAC, we propose an efficient multiplier circuit with high accuracy and performance in Section III-B.

B. Approximate 8-bit Multiplier

We show the schematic of the proposed multiplier in Fig. 3 leveraging the proposed ACMLC-based compressor and CAC. Our proposed approximate multiplier comprises three components, truncation, constant truncation, approximation, and exact compute, each contributing to the computation of partial products (PPs) in Stage 1 (c.f., Fig. 3). One of the common methods in these circuits is to truncate at least four LSB columns of PPs (for an 8-bit multiplier). This approach is

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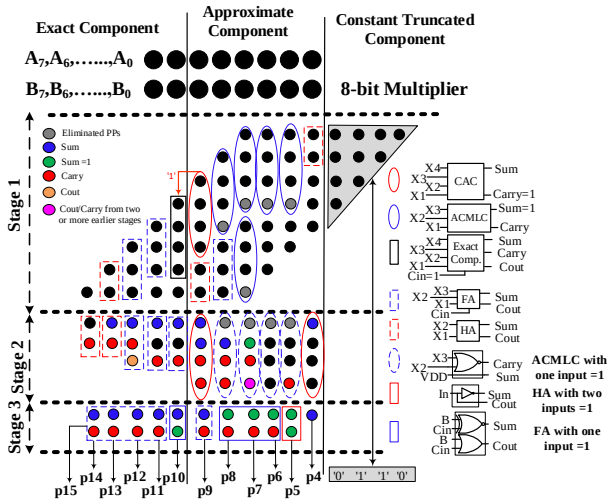

Fig. 3. Proposed multiplier schematic using the proposed ACMLC-based compressor and CAC.

TABLE II
PERFORMANCE OF 4:2 COMPRESSORS.

Circuit Architectures	Power (μW)	Delay (ps)	PDP (fJ)	Transistors	ADP (ns × #T)	PAP (μW × #T)	PDAP (fJ × #T)
Exact [15]	2.68	745.48	2	38	28.32	101.98	76.02
TC'15 [3]	1.72	248.73	0.427	26	6.46	44.72	11.12
TVLSI'17 [4]	2.74	238.05	0.653	36	8.57	98.85	23.52
ESL'18 [5]	3.2	233.21	0.747	30	6.99	96.09	22.41
TVLSI'17 [6]	2.04	240.11	0.491	26	6.24	53.26	12.78
TCAS1'19 [7]	0.75	239.88	0.180	12	2.87	9.04	2.16
IJEC'19 [8]	0.7	250.84	0.175	16	4.01	11.2	2.8
TCAS-1'20 [9]	2.97	268.38	0.797	46	12.34	136.65	36.67
ESL'23 [11]	1.63	243.84	0.399	48	11.70	78.55	19.15
TCAS-2'23 [12]	0.7	258.03	0.180	26	6.7	18.2	4.69
ACMLC (this work)	0.61	258.03	0.158	8	2.06	4.9	1.26
CAC (this work)	0.64	247.06	0.158	14	3.45	9	2.22

* Best and worst results are highlighted in green and red, respectively.

effective with highly accurate approximate components. To mitigate a higher number of errors with the proposed compressors, an alternative method is preferred in this work based on [13]. In [13], constant values are assigned to several LSB PP bits. The probabilities of generating 0 or 1 for each of the circuits generating p0, p1, p2, p3, and p4 were investigated by applying different inputs. As per observations, the rightmost four LSB bits are fixed at p0p1p2p3=0110. By fixing the LSB bits to a constant value, the number of required gates is reduced. In this case, ten AND gates are removed from the first four columns of the proposed multiplier circuit (as shown with the gray triangular shaded area in Fig. 3).

The approximate component of the proposed multiplier comprises five ACMLC-based compressors, a CAC, two exact HAs and an exact FA. With the proposed 3-input ACMLC-based compressor, one of the four PPs is effectively eliminated, resulting in the elimination of five AND gates within a single PP stage. The eight unused (solid black circles) AND gates in that stage are repurposed in the second stage to form a compressor-based chain, thereby reducing the overall area. The proposed CAC is used in the last column of the approximate component to transfer $Carry = 1$ to the exact component, simplifying the exact compressor circuit. Finally, the simplified exact compressor [15] (with constant input, $Cin = 1$) is only utilized once in the entire multiplier. Note that the use of the exact HAs and FA enahnces multiplier accuracy, complementing the small area of the proposed ACMLC-based compressor and CAC. We describe resit of the circuit's function below.

TABLE III
PERFORMANCE OF MULTIPLIERS.

Circuit Architectures	Power (μW)	Delay (ps)	PDP (fJ)	Transistors	Norm. PDAP	NMED $\times 10^3$	MRED $\times 10^{-5}$
Exact	29.07	252.16	7.33	1,530	1	—	—
[3]	6.21	228.02	1.41	1,202	84%	1.6	10.8
[4]	7.25	237.54	1.72	1,322	79%	1.52	9.43
[5]	7.45	226.49	1.68	1,250	81%	1.46	9.51
[6]	6.38	232.32	1.48	1,202	84%	1.37	9.26
[7]	3.57	234.49	0.83	900	93%	1.61	10.8
[8]	5.11	238.89	1.22	1,082	88%	1.42	9.54
[9]	7.03	238.54	1.67	1,442	78%	1.3	9.46
[11]	5.56	254.39	1.41	1,466	81%	1.47	9.63
[12]	5.84	233.87	1.36	1,202	85%	1.47	9.63
[16]	2.45	238.33	0.58	698	96.3%	1.58	9.95
Proposed	2.16	236.45	0.51	786	96.4%	1.71	10.7

* Best and worst results are highlighted in green and red, respectively.

TABLE IV
APPROXIMATE MULTIPLIERS PSNR RESULTS IN IMAGE MULTIPLICATION.

Circuit Architectures	Cameraman × Circle (dB)	Cameraman × Rice (dB)	Lenna × Circle (dB)	Average PSNR (dB)	OA (%)	FoM OA/PDP(%/fJ)
[3]	40.31	41.18	40.12	40.54	91	64.54
[4]	39.42	40.12	40.11	39.88	90	52.33
[5]	38.89	40.84	39.67	39.8	92	54.76
[6]	40.21	39.85	39.85	39.97	93	62.84
[7]	39.67	39.59	39.34	39.53	91	109.64
[8]	40.02	39.35	40.61	39.99	93	76.23
[9]	40.41	40.03	41.43	40.62	92	55.09
[11]	39.42	39.11	40.72	39.75	92	65.25
[12]	39.42	40.03	39.5	39.65	92	67.65
[16]	40.22	40.12	39.92	40.09	90	155.17
Proposed	40.24	40.03	40.31	40.19	91	178.43

In Stage 2, the first and last columns of the approximate component consist of CAC. As a result, more AND gates are utilized, exhibiting higher compression accuracy. The proposed ACMLC-based Stage 1 architecture enables utilization of an additional four ACMLC-based compressors in Stage 2 where input X4 highlighted in with gray circles in Fig. 3, are ignored. As a result, 15 out of the 64 AND gates (as required for the 8-bit multiplier), are removed, requiring only 49 gates with the proposed architecture. Additionally, Stage 2 uses two exact HAs and three exact FAs. The $Carry$ output from the CAC in Stage 2 is transferred to the final addition level.

Stage 3 typically uses varying bit-length ripple carry adders (RCAs) requiring a large number of HAs and FAs. ACMLC-based compressor and CAC with $Sum = 1$ and $Carry = 1$, respectively, can significantly reduce this number. Therefore, the approximate component in the final stage employs an HA with two constant inputs of '1', functioning as an inverter. Furthermore, the three FAs with a constant input of '1' can be implemented using an XOR and an OR (as shown in Fig. 3).

Intuitively, the overall simplicity of our architecture as compared with an exact multiplier or the ML-based multiplier [7] is expected to yield significant savings in power, area, and delay. Our experimental results in Section IV show that the proposed multiplier comprising 786 transistors is 49% smaller than the 1530-transistor exact multiplier [15] and 12.6% smaller than the 900-transistor ML-based multiplier [7].

IV. EXPERIMENTAL SETUP AND RESULTS

We investigate and compare our compressors and multiplier with the prior work reviewed in Section II. Please note that all the circuit performance results, including power consumption, delay time, and power-delay product (PDP) are reported based on HSPICE simulation with 7-nm FinFET technology under equal conditions [7]. For the delay, the worst-case condition has been reported when the input and output signals approach 50% of the supply voltage. The normalized mean error

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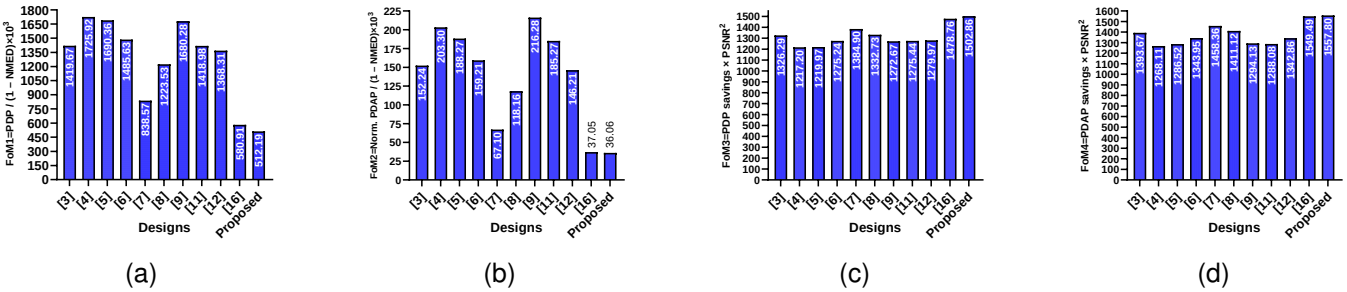

Fig. 4. Image multiplication performance with approximate multipliers with respect to FoMs – (a): FoM1, (b): FoM2, (c): FoM3, (d): FoM4.

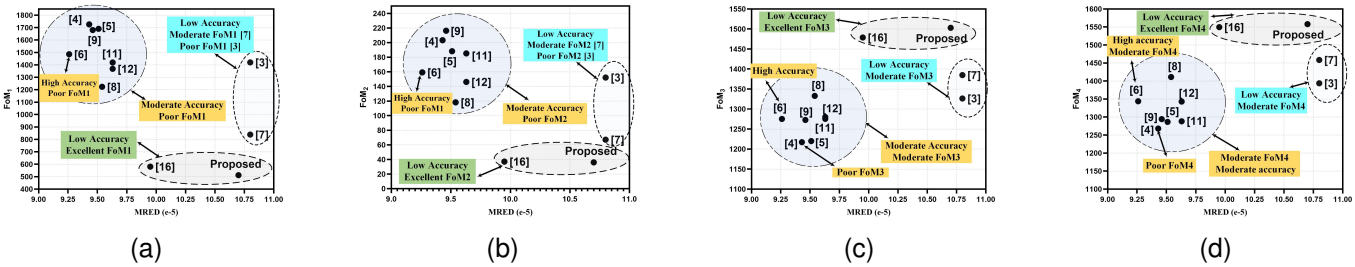

Fig. 5. Performance of the multipliers with respect to FoMs 1 - 4 vs. MRED.

distance (NMED) and mean relative error distance (MRED) accuracy metrics are reported based on MATLAB simulation. The operation is set at a supply voltage of 0.7V and a frequency of 2GHz. The FinFET model parameters are based on [7], except for the Fin height and gate length, which are set to 18nm and 11nm, respectively, for both p-type and n-type transistors.

We evaluate precision in MATLAB using one million random input combinations based on NMED and MRED [7]. We report simulation results with respect to worst-case delay, power, and PDP. Additionaly, we report area-based metrics, *e.g.*, area-delay product (ADP), power-area product (PAP), and power-delay-area product (PDAP) based on the transistor numbers. Also, we consider various Figures of Merits (FoMs) with distinct characteristics to facilitate comparisons based on circuit performance and accuracy. We showcase our proposed circuits using image multiplication as the target use case.

A. Approximate Compressors

Table II shows the performance of the proposed and existing compressors. With only eight transistors, ACMLC is the smallest among the compared circuits. With low transistor count and low parasitic capacitance, the ACMLC compressor exhibits the lowest power consumption among all the evaluated circuits. The proposed 14-transistor CAC ranks as the second lowest in power consumption among all the circuits. Although the ACMLC and CAC are ∼10% slower than [5], [7], they exhibit the lowest PDP due to the significant savings in power. With respect to ADP, PAP, and PDAP, the ACMLC-based compressor, ML-based compressor [7], and CAC exhibit improvements of 95.33%, 92.66%, and 92%, respectively, compared to the exact compressor [15]. On average, the ACMLC demonstrates superior performance compared to prior art, with the proposed CAC ranking third, just after [7].

B. Approximate Multipliers

Table III indicates the performance comparison among various multipliers. After the multiplier of [16], which is the closest competitor to suggested multiplier, the proposed 786-transistor multiplier is the second smallest circuit, with a 49% area reduction compared to the exact multiplier. The proposed multiplier also exhibits superior PDP (0.51 fJ) and PDAP (96% improvement compared to the exact circuit) performance metrics. The superior performance of the proposed multiplier is due to the underlying ACMLC-based compressor and CAC, as well as the multiplier architecture tailored to exploit the advantages of the proposed compressors. As expected, the proposed multiplier does not exhibit superior accuracy compared to state-of-the-art solutions shown in Table III. Nevertheless, the error rate falls within the typical range for approximate multipliers and is lower than the error rate of the baseline ML-based compressor in terms of power, area, and delay performance (c.f., Table II). The improved accuracy of the proposed multiplier is attributed to a reduced number of negative errors and cumulative negative EDs compared to [7].

C. Image Multiplication

Multiplication is crucial in image processing tasks, demanding high computational resources. Therefore, efficient approximate multipliers are crucial for effective and faster computation with acceptable precision. To evaluate the performance accuracy, we evaluated our multiplier using image multiplication in MATLAB. The quality of the output image is determined based on the peak signal-to-noise ratio (PSNR) with the sets of images listed in Table IV. The quality of image multiplication remains consistent across all evaluated circuits, as evidenced by the PSNR values, ranging between 39.54 dB and 40.63 dB. To further investigate the accuracy of the proposed circuits in practical applications, the performance of the circuits is evaluated with respect to quality of the produced images using the overall accuracy (OA) for various sets of

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images, as listed in Table IV. The OA metric is computed as (TP+TN)/(total number of samples), where TP and TN are, respectively, the true positives and true negatives, and are computed in MATLAB [17]. A figure of merit (FoM) which considers both the circuit accuracy, and power and delay performance (FoM = OA/PDP [%/fJ]) is also shown in Table IV. While the OA for the proposed circuit is comparable to the OA of state-of-the-art approximate multipliers, the FoM for the proposed solution is significantly higher than prior art, exhibiting higher accuracy per unit energy. Note that the proposed circuit yields 91% OA with a 13% and 39% higher FoM than the second [16] and third [7] best solutions.

We consider different FoMs to further evaluate the proposed multiplier with image multiplication as shown in Fig. 4. Lower values of FoM1 and FoM2 and higher values of FoM3 and FoM4 indicate better performance of multipliers. Due to the excellent power and area characteristics shown by the PDP and PDAP metrics and considering the typical multiplication quality determined by the NMED metric, our multiplier demonstrates superior performance across all four FoMs.

As compared with the ML-based multiplier [7], the ACMLC/CAC multiplier yields 38.92% (FoM1), 46.2% (FoM2), 7.84% (FoM3), and 6.38% (FoM4) better performance as shown in Fig. 4. However, compared to [16], the proposed cell has a slight superiority. According to Fig. 5, the highly accurate circuits exhibit poor FoMs, while our lower-accuracy multipliers, [3], and [7] yield superior FoM performance. Note that the proposed multiplier outperforms [3] and [7] regarding MRED and FoMs. The primary goal is system performance in error-resilient applications (*i.e.*, power efficiency and speed) at minimum sufficient accuracy within a small footprint. Based on the Pareto chart in Fig. 5, the proposed circuits exhibit lower accuracy than the multipliers in [4]- [6] [8] [9] [11] [12] [16]. Yet, the error with the proposed multipliers is sufficiently small to yield excellent performance in the target error-resilient ML applications (see PSNR and OA in image multiplication as shown in Table IV). Alternatively, the proposed multiplier yields a significantly better PDP and OA/PDP than the prior art (Fig. 4 and Table IV). Although the circuit in [16] is competitive from various perspectives, the proposed circuit outperforms it in all considered FoMs.

V. Conclusion

In this paper, we proposed an 8-transistor ACMLC compressor, a 14-transistor CAC, and an approximate 8-bit multiplier for accurate and efficient image multiplication. The compressor has small footprint and low power consumption at the expense of a relatively high error rate. To compensate for the negative errors, we propose CAC, exhibiting seven errors, with only one being negative. We propose an ACMLC/CAC-based approximate multiplier to exploit the proposed compressors' unique characteristics. Relative to an exact multiplier, the proposed multiplier exhibits 50% area reduction and 93% power savings. The proposed multiplier exhibits superior performance across most evaluated metrics compared to state-of-the-art approximate multipliers. The Pareto results reveal that despite their lower accuracy, ML-based proposed circuits are promising for low-power and energy dissipation applications.

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